

Random maths

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0.1	October 7, 2026	First draft

Abstract

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1 Curvature

Table 1: Curve definitions used in this section.

Term	Definition	Example
Space curve	This is the most generic definition of a curve. A geometric path $\gamma(\lambda)$ in $\mathbb{R}^D$ with an arbitrary parameter λ . It describes only the shape, not how fast it is traversed.	$\gamma_1(\lambda) = (\cos 3\lambda, \sin 3\lambda)$ has speed $\frac{d\gamma}{d\lambda} = 3$ everywhere.
Trajectory	A space curve parametrized by time, $r(t) = \gamma(s(t))$, the position of the robot at time t .	$r(t) = \gamma_1(t) = (\cos 3t, \sin 3t)$ is the unit circle traversed at constant speed $v = 3$.
Unit-speed curve	A space curve parametrized by arc length s , so that $\left\ \frac{d\gamma(s)}{ds} \right\ = 1$ for all s . We can define any space curve by a unit-speed curve.	$\gamma_2(\lambda) = (\cos \lambda, \sin \lambda)$ has speed 1 everywhere. It is the same curve as γ_1 , we just normalized it.

1.1 Relation between a unit-speed curve and a trajectory in the time domain

A unit-speed curve $\gamma(s)$ is parametrized by arc length s . Formally, we define a unit-speed curve by:

$$\left\| \frac{d\gamma(s)}{ds} \right\| = 1, \quad \forall s, \quad (1)$$

$$\mathbf{T}(s) = \frac{d\gamma(s)}{ds}, \quad (2)$$

$$\kappa(s) = \left\| \frac{d\mathbf{T}(s)}{ds} \right\|, \quad (3)$$

$$\mathbf{N}(s) = \frac{1}{\kappa(s)} \frac{d\mathbf{T}(s)}{ds}, \quad \kappa(s) > 0. \quad (4)$$

Here, $\mathbf{T}$ is the unit tangent vector to the curve. Since $\mathbf{T}$ has constant length, it can only rotate, so $\frac{d\mathbf{T}}{ds}$ is perpendicular to $\mathbf{T}$ and $\mathbf{N}$ is a unit vector perpendicular to $\mathbf{T}$. The curvature κ is a magnitude, $\mathbf{N}$ is its direction. Intuitively, curvature is how much a curve deviates from being a straight line [1].

A trajectory $r(t)$ is the position of the robot as a function of time. We can rewrite it as $r(t) = \gamma(s(t))$ where $s(t)$ is the arc-length profile. The speed is the rate of change of the arc length,

$$v(t) = \frac{ds}{dt}. \quad (5)$$

A unit-speed curve is traversed at constant speed, so a change of acceleration can only change the heading. By adding time here, the speed is no longer constant and we have a new component of the acceleration that is tangent to the curve. Tangential acceleration modifies the speed and perpendicular (normal) acceleration modifies the heading. Formally, using the chain rule and the product rule, we have:

$$\frac{dr(t)}{dt} = \frac{d\gamma(s)}{ds} \frac{ds}{dt} = v \mathbf{T}(s(t)), \quad (6)$$

$$\frac{d^2r(t)}{dt^2} = \frac{dv}{dt} \mathbf{T} + v \frac{d\mathbf{T}(s)}{ds} \frac{ds}{dt} = \frac{dv}{dt} \mathbf{T} + \kappa v^2 \mathbf{N}. \quad (7)$$

The term $\frac{dv}{dt} \mathbf{T}$ is the tangential acceleration and $\kappa v^2 \mathbf{N}$ is the centripetal acceleration, which vanishes on a straight line ($\kappa = 0$).

1.2 Computing curvature from a discretized space curve

To compute curvature from a discretized space curve, we need to derivate. An easy way to do it is a central finite difference. With ϵ_1 and ϵ_2 the distances from s_i to its two neighbors, the curvature is the inverse of the radius r_i of the circle centered at the ICR. Note that we do not depend on time

here and if we are using a trajectory we need to reparametrize it to distance traveled. Here is the central difference curvature computation:

$$\kappa_i = \frac{1}{r_i} = \frac{2\Delta\theta}{\epsilon_1 + \epsilon_2}, \quad (8)$$

where $\Delta\theta$ is the heading change at s_i . By choosing a small $\epsilon_{\min}$ we can bump into noise. Conversely, choosing a big $\epsilon_{\min}$ will miss rapid changes of curvature.

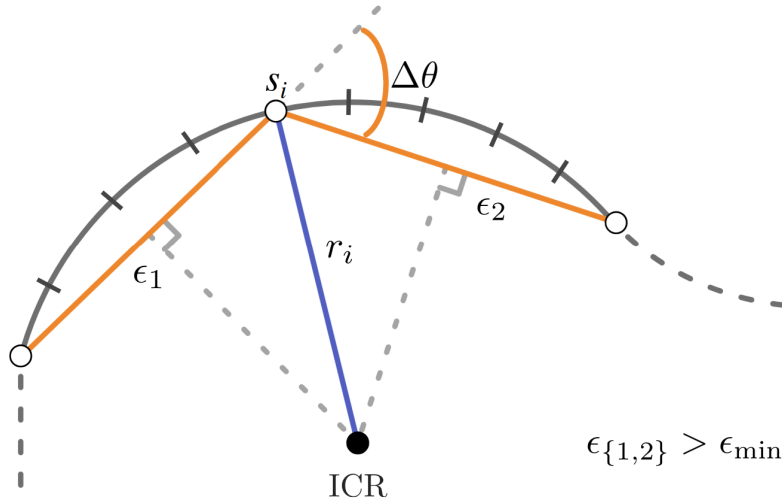


Figure 1: Curvature at a point s_i of a discretized space curve. The ICR is the center of the circle through the neighboring points, r_i is its radius and $\Delta\theta$ is the heading change at s_i . The neighbors are chosen at least $\epsilon_{\min}$ away from s_i . Figure from [2]

2 Inertial Measurement Unit

An Inertial Measurement Unit (IMU) is composed of a tri-axial gyroscope and a tri-axial accelerometer. The gyroscope measures angular speed and the accelerometer measures linear accelerations. Proper acceleration is the acceleration of an object relative to freefall. An accelerometer measures proper acceleration, meaning that it senses gravitational acceleration at rest. An Inertial frame is a frame that has constant speed. In robotics, the body frame $\mathcal{B}$ is Non-Inertial and the global frame $\mathcal{G}$ is an Inertial frame. We can call earth an Inertial frame if we ignore the fact it rotates. Fictitious forces (centrifugal, Euler and Coriolis) appear when we try to explain the movement of an object in a Non-Inertial frame of reference. We only consider fictitious forces from earth rotation for long term Inertial navigation.

2.1 Measurement model

We model the IMU gyroscope and accelerometer measurement ($\tilde{\mathbf{a}}$ and $\tilde{\boldsymbol{\omega}}$) as follows:

$${}^{\mathcal{B}}\tilde{\mathbf{a}} = {}^{\mathcal{B}}\mathbf{a} - {}^{\mathcal{B}}\mathbf{R}^{\mathcal{G}}\mathbf{g} + \boldsymbol{\eta}^{\mathbf{a}}(t) + \mathbf{b}^{\mathbf{a}}(t) \quad (9)$$

$${}^{\mathcal{B}}\tilde{\boldsymbol{\omega}} = {}^{\mathcal{B}}\boldsymbol{\omega} + \boldsymbol{\eta}^{\boldsymbol{\omega}}(t) + \mathbf{b}^{\boldsymbol{\omega}}(t) \quad (10)$$

$\mathcal{B}$ and $\mathcal{G}$ are respectively the body and global frame. Both gyroscope and accelerometer are corrupted by a slowly varying bias $\mathbf{b}$ and noise $\boldsymbol{\eta}$.

2.2 Noise model

An Allan variance plot (Fig. 2) is the best way to explain the noises of an IMU. The x axis τ is the window length used to compute the statistics. The y axis is the standard deviation over all the time windows. Intuitively, we can read the y as how much the average measurement moves between windows. The first downslope is where white noise dominates. When we average on a small window, we get a large variance caused by white noise and as we average more and more values we are able to minimize the impact of white noise and lower the variance up to a floor. That floor is the bias (in) stability and it represents how precise an IMU can ever be. The upslope is the bias random walk. On the upslope, bias varies enough inside a measurement window that we raise the variance as we pick longer time window.

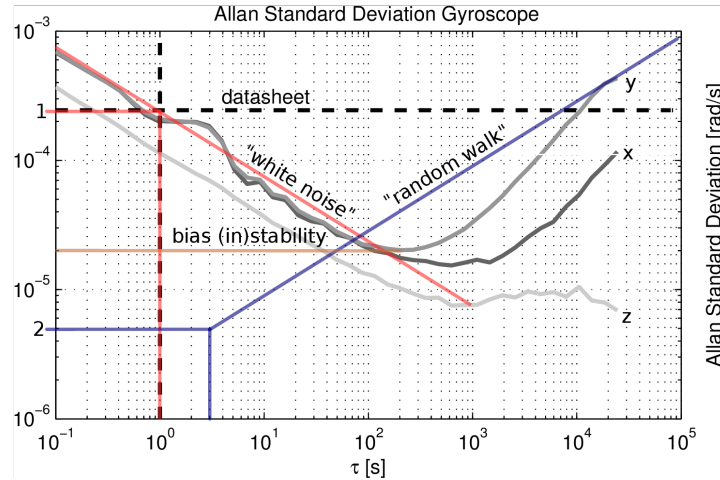


Figure 2: Allan standard deviation of a gyroscope for the x , y and z axes. The white noise and random walk lines are fitted to the curves, and the bias instability is the minimum of the curve. The red lines intersection is the noise density and the blue one is the random walk. Noise density and random walk values are used in state estimators. Figure from [3].

In a state estimator, we use the noise density and random walk values that respectively quantify the white noise magnitude and how fast our bias moves over time. The bias (in) stability floor

is how we can quantify how precise our IMU can ever be. An IMU is a high frequency sensor that could yield high accuracy trajectory from numerical integration if we were able to compensate perfectly for the noise and bias. We typically use them as a prior for an exteroceptive sensor that does the heavy lifting. I like the philosophy behind Zhao *et al.* [4] where the narrative is switched : the exteroceptive sensor act as constraints on the IMU noise and bias allowing it to yield accurate trajectories.

3 State relations

The states of the robot are linked by three kinematic equations,

$${}^{\mathcal{G}}\dot{\mathbf{p}} = {}^{\mathcal{B}}\mathbf{R}^T {}^{\mathcal{B}}\mathbf{v}, \quad (11)$$

$${}^{\mathcal{B}}\dot{\mathbf{v}} = {}^{\mathcal{B}}\mathbf{a} - {}^{\mathcal{B}}\boldsymbol{\omega} \times {}^{\mathcal{B}}\mathbf{v}, \quad (12)$$

$${}^{\mathcal{B}}\dot{\mathbf{R}} = -{}^{\mathcal{B}}\boldsymbol{\omega}^{\wedge} {}^{\mathcal{B}}\mathbf{R}, \quad (13)$$

where $(\cdot)^{\wedge}$ builds the skew-symmetric matrix such that $\mathbf{a}^{\wedge}\mathbf{b} = \mathbf{a} \times \mathbf{b}$. Left superscripts give the frame, global $\mathcal{G}$ or body $\mathcal{B}$, and ${}^{\mathcal{B}}\mathbf{R}$ rotates vectors from $\mathcal{G}$ to $\mathcal{B}$. In Eq. (12), ${}^{\mathcal{B}}\dot{\mathbf{v}}$ is the rate of change of the components of ${}^{\mathcal{B}}\mathbf{v}$ in $\mathcal{B}$; the term ${}^{\mathcal{B}}\boldsymbol{\omega} \times {}^{\mathcal{B}}\mathbf{v}$ accounts for the rotation of the body frame. These equations hold in 3D for any rigid body, with or without slip.

Figure 3 shows which quantities are needed to move from one state to another. Each arrow lists all the quantities needed to compute the target state; integrating (up) also needs an initial value of the target state.

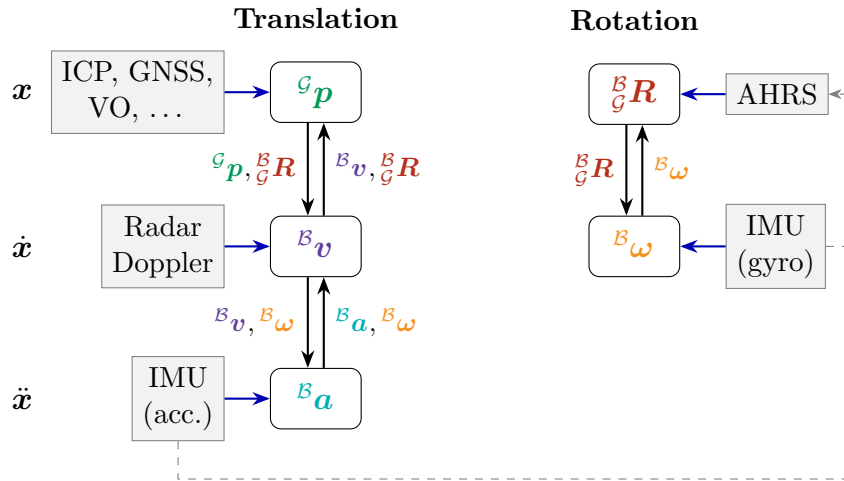


Figure 3: States of the robot. Down arrows differentiate and up arrows integrate following Eqs. (11) to (13), and their labels give the quantities needed. Blue arrows are direct measurements and dashed arrows are dependencies.

3.0.1 Discretization

Integrating Eqs. (11) to (13) with explicit Euler over a time step Δt gives the updates on the left. Solving them for the lower-order state gives the forward differences on the right, so each pair is an exact inverse.

$$\begin{aligned}
 {}^{\mathcal{G}}\mathbf{p}_{t+\Delta t} &= {}^{\mathcal{G}}\mathbf{p}_t + {}^{\mathcal{B}}\mathbf{R}_t^T {}^{\mathcal{B}}\mathbf{v}_t \Delta t & {}^{\mathcal{B}}\mathbf{v}_t &= {}^{\mathcal{B}}\mathbf{R}_t \frac{{}^{\mathcal{G}}\mathbf{p}_{t+\Delta t} - {}^{\mathcal{G}}\mathbf{p}_t}{\Delta t} \\
 {}^{\mathcal{B}}\mathbf{v}_{t+\Delta t} &= \text{Exp}(-{}^{\mathcal{B}}\boldsymbol{\omega}_t \Delta t) ({}^{\mathcal{B}}\mathbf{v}_t + {}^{\mathcal{B}}\mathbf{a}_t \Delta t) & {}^{\mathcal{B}}\mathbf{a}_t &= \frac{\text{Exp}({}^{\mathcal{B}}\boldsymbol{\omega}_t \Delta t) {}^{\mathcal{B}}\mathbf{v}_{t+\Delta t} - {}^{\mathcal{B}}\mathbf{v}_t}{\Delta t} \\
 {}^{\mathcal{B}}\mathbf{R}_{t+\Delta t} &= \text{Exp}(-{}^{\mathcal{B}}\boldsymbol{\omega}_t \Delta t) {}^{\mathcal{B}}\mathbf{R}_t & {}^{\mathcal{B}}\boldsymbol{\omega}_t &= -\frac{\text{Log}({}^{\mathcal{B}}\mathbf{R}_{t+\Delta t} {}^{\mathcal{B}}\mathbf{R}_t^T)}{\Delta t}
 \end{aligned}$$

$\text{Exp}(\cdot)$ maps a rotation vector to $SO(3)$ and $\text{Log}(\cdot)$ is its inverse. The position and velocity updates are first order in Δt . The rotation update is exact when ${}^{\mathcal{B}}\boldsymbol{\omega}$ is constant over Δt , as with gyroscope samples. The forward difference at t is the same value as a backward difference at $t + \Delta t$; only the timestamp differs, by half a step from the true derivative at $t + \Delta t/2$.

Curvature

$$\begin{aligned}
 \kappa &= \frac{\Delta\theta}{\Delta s} \\
 \kappa &= \frac{1}{r}
 \end{aligned}$$

Accelerometer model

$$\begin{aligned}
 {}^{\mathcal{B}}\tilde{\mathbf{a}} &= {}^{\mathcal{B}}\mathbf{a} + {}^{\mathcal{B}}\mathbf{R}^{\mathcal{G}}\mathbf{g} + \boldsymbol{\eta}(t) + \mathbf{b}(t) \\
 {}^{\mathcal{B}}\mathbf{a} &= {}^{\mathcal{B}}\mathbf{a}_c + \dots \\
 \mathbf{a}_c &= \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') \\
 F &= ma
 \end{aligned}$$

$\mathbf{r}'$ is the position vector of the object relative to the rotating reference frame [5].

Links using scalars

$$\begin{aligned}
 a_c &= \frac{v^2}{r} \\
 v &= \omega r \\
 v &= \frac{\omega}{\kappa}
 \end{aligned}$$

References

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